THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A die is thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. The experimental probability of getting 2 is: 1

(a) $\frac{11}{51}$ **(b)** $\frac{6}{25}$ **(c)** $\frac{39}{50}$ **(d)** $\frac{11}{50}$

2. In a survey of 500 families, 140 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{7}{26}$ **(b)** $\frac{18}{25}$ **(c)** $\frac{7}{25}$ **(d)** $\frac{8}{25}$

3. A letter is chosen at random from the word EDUCATION. The probability that it is a vowel is: 1

(a) $\frac{4}{9}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{5}{9}$ **(d)** $\frac{2}{3}$

4. Three coins are tossed together. The probability of getting exactly one head is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{3}{8}$

5. A bag has 4 red, 3 black and 5 white balls. The probability of drawing a ball that is not black is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{1}{4}$

6. If $P(E) = 0.3$, then $P(\text{not } E) \times 100$ equals: 1

(a) 69 **(b)** 70 **(c)** 72 **(d)** 71

7. A fair die is thrown once. The probability of getting a composite number is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{6}$

8. The probability of an event E always satisfies: 1

(a) $P(E) > 1$ **(b)** $0 \le P(E) \le 1$ **(c)** $-1 \le P(E) \le 1$ **(d)** $P(E) < 0$

9. A die is thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. The experimental probability of getting 5 is: 1

(a) $\frac{7}{51}$ **(b)** $\frac{7}{50}$ **(c)** $\frac{4}{25}$ **(d)** $\frac{43}{50}$

10. In a survey of 400 families, 90 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{9}{40}$ **(c)** $\frac{31}{40}$ **(d)** $\frac{9}{41}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A die was thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. Find the experimental probability of getting an even number. 2

12. A letter is chosen at random from the word MATHEMATICS. Find the probability that it is a consonant. 2

13. Three coins are tossed together. Find the probability of getting exactly one head. 2

14. A die was thrown 100 times with frequencies of $1, \ldots, 6$ being 18, 12, 20, 15, 19, 16. Find the experimental probability of getting an even number. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains 4 red, 3 black and 5 white balls. A ball is drawn at random. Find the probability that it is — 3
(i) red,
(ii) not red,
(iii) black or white.

16. In a survey of 200 students, 80 take the bus, 50 cycle and the rest walk. Find the probability that a randomly chosen student — 3
(i) walks,
(ii) does not take the bus.

17. Two coins are tossed 100 times: two heads occurred 24 times, one head 52 times and no head 24 times. Find the experimental probability of each of the three events. — 3

OR

A bag contains 6 red, 2 black and 7 white balls. A ball is drawn at random. Find the probability that it is — 3
(i) red,
(ii) not red,
(iii) black or white.

18. In a survey of 250 students, 100 take the bus, 75 cycle and the rest walk. Find the probability that a randomly chosen student — 3
(i) walks,
(ii) does not take the bus.

OR

Two coins are tossed 200 times: two heads occurred 45 times, one head 105 times and no head 50 times. Find the experimental probability of each of the three events. — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A die was thrown 100 times; the number of times $1, \ldots, 6$ appeared was 15, 22, 17, 16, 14, 16. Find the experimental probability of — 5
(i) a prime number,
(ii) a number greater than 4,
(iii) a number less than 3, and check that all six relative frequencies add to 1.

OR

A letter is chosen at random from the word MISSISSIPPI. Find the probability that it is — 5
(i) a vowel,
(ii) the letter S,
(iii) a letter from the first half of the alphabet (A to M).

20. A die was thrown 100 times; the number of times $1, \ldots, 6$ appeared was 18, 12, 20, 15, 19, 16. Find the experimental probability of — 5
(i) a prime number,
(ii) a number greater than 4,
(iii) a number less than 3, and check that all six relative frequencies add to 1.

OR

A letter is chosen at random from the word MATHEMATICS. Find the probability that it is — 5
(i) a vowel,
(ii) the letter M,
(iii) a letter from the first half of the alphabet (A to M).

END OF QUESTION PAPER — Total: 40 Marks